Yoshiki Nakamura
Naosuke Matsuda

On Implicational Intermediate Logics Axiomatizable by Formulas Minimal in Classical Logic: A Counter-Example to the Komori–Kashima Problem

Abstract. The Komori–Kashima problem, that asks whether (or not) the implicational intermediate logics axiomatizable by formulas minimal in classical logic are only intuitionistic logic and classical logic, has stood for over a decade. In this paper, we give a counter-example to this problem. Additionally, we also give some open problems derived from this result.

Keywords: Intermediate logic, Minimal formulas, Axiomatization.

1. Introduction

In [9], Komori investigated some logics (including implicational intermediate logics) through the substitution preorder $\leqslant_{\mathrm{sub}}$, which is defined as follows: for two formulas, α and β,

$$\alpha \leqslant_{\mathrm{sub}} \beta \text{ iff } \beta \text{ is a substitution instance of } \alpha.$$

Long afterwards, on implicational intermediate logics axiomatizable by formulas minimal in classical logic w.r.t. the substitution preorder $\leqslant_{\mathrm{sub}}$, Komori [10] and Kashima [8] posed the following problem in 2005.

Problem. (the Komori–Kashima problem) The implicational intermediate logics axiomatizable by formulas minimal in classical logic (w.r.t. $\leqslant_{\mathrm{sub}}$) are only intuitionistic logic (IL) and classical logic (CL)?[1]

[1]The original problem posed by Komori [10, 1.] [12, Conjecture 4.1] is stronger as follows: "how many logics L including **BCK** are there such that L is axiomatizable by formulas minimal in classical logic? Such logics are only **BCK**, IL, and CL?".

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Kashima [8] introduced a sequent calculus towards solving this problem positively and presented a partial positive solution to the problem. However, the problem has remained open for over a decade.

In this paper, we solve this problem negatively. We show that there exists some implicational intermediate logic neither IL nor CL such that it is axiomatizable by formulas minimal in CL. More concretely, we construct a formula minimal in CL such that the logic axiomatized by the formula is equivalent to the logic LC (a.k.a. Gödel-Dummett logic [2]). Here, LC is the intermediate logic axiomatized by the following formula $\mathbf{G}$: [2]

$$\mathbf{G} \triangleq ((a \to b) \to c) \to ((b \to a) \to c) \to c.$$

The formula $\mathbf{G}$ is not minimal in CL (Example 2.4) as it is, but we can obtain a formula $\mathbf{G}'$ such that $\mathbf{G}'$ is minimal in CL and $\mathrm{IL} + \mathbf{G}' = \mathrm{IL} + \mathbf{G}$, fortunately (Sect. 4). Hence, the Komori–Kashima problem is negatively solved.

Remark 1.1. The Komori–Kashima problem is for (purely) implicational formulas. For full formulas, it was known that there exist counter-examples. Kashima [7] found that the formula $\mathbf{J} \triangleq \neg(a \wedge b) \to (\neg a \vee \neg b)$, which defines Jankov-De Morgan logic (KC) [6], is minimal in CL. Afterwards, Hirai [4] presented the formula $\mathbf{J}' \triangleq (a \to b) \vee ((a \to \bot) \to c)$, which is also minimal in CL and defines the logic KC. Furthermore, the formula $\mathbf{J}'' \triangleq ((a \to b) \to d) \to (((a \to \bot) \to c) \to d) \to d$, which is the formula obtained from $\mathbf{J}'$ by destructing $\vee$, is also minimal in CL and $\mathrm{IL} + \mathbf{J}'' = \mathrm{KC}$. However, $\mathbf{J}$, $\mathbf{J}'$, and $\mathbf{J}''$ are not implicational formulas, since they use the negation symbol $\neg$ or the bottom symbol $\bot$.

Outline Section 2 presents the preliminary definitions and Section 3 states the Komori–Kashima problem and presents a few observations for the problem. Section 4 presents a counter-example formula to the problem. Section 5 concludes this paper and presents some open problems.

2. Preliminaries

Given a countable set PV of *propositional variables*, we define the set Fml of *(implicational) formulas* by:

$$\mathrm{Fml} ::= \mathrm{PV} \mid (\mathrm{Fml} \to \mathrm{Fml}).$$

[2] $\mathbf{G}$ is obtained from the formula $(a \to b) \vee (b \to a)$ (so-called *prelinearity axiom*) by destructing $\vee$.

We use letters $a, b, c, \ldots, z$ to stand for arbitrary propositional variables, and use $\alpha, \beta, \gamma, \delta$ for arbitrary formulas. Parentheses may be omitted in such a way that for example, $\alpha \to \beta \to \gamma$ denotes the formula $(\alpha \to (\beta \to \gamma))$. $[\beta/p]\alpha$ stands for the formula obtained from α by replacing each occurrence of p by β.

Logic A set $L \subseteq \mathrm{Fml}$ is called a *logic* if L satisfies the following:

- if $\alpha \to \beta \in L$ and $\alpha \in L$, then $\beta \in L$; and
- if $\alpha \in L$, then $[\beta/p]\alpha \in L$ for each $\beta \in \mathrm{Fml}$ and $p \in \mathrm{PV}$.

For a set $F \subseteq \mathrm{Fml}$, we write F^* for the smallest logic including F. We say that L is *axiomatized* (or *defined*) by F if $L = F^*$ holds. For simplicity, we abbreviate $(L \cup F)^*$ as $L + F$ for a set F of formulas, and abbreviate $L + \{\alpha\}$ as $L + \alpha$ for a formula α.

Intermediate Logic Intuitionistic logic (IL) and classical logic (CL) are the logics $\{\mathbf{K}, \mathbf{S}\}^*$ and $\{\mathbf{K}, \mathbf{S}, \mathbf{P}\}^*$ (so, $\mathrm{CL} = \mathrm{IL} + \mathbf{P}$), where $\mathbf{K} \triangleq a \to b \to a$, $\mathbf{S} \triangleq (a \to b \to c) \to (a \to b) \to a \to c$, and $\mathbf{P} \triangleq ((a \to b) \to a) \to a$. An *intermediate logic* is a logic L such that $\mathrm{IL} \subseteq L \subseteq \mathrm{CL}$. In this paper, we say that an intermediate logic L is *proper* if $\mathrm{IL} \subsetneq L \subsetneq \mathrm{CL}$.

For example, $\mathrm{LC} = \{\mathbf{K}, \mathbf{S}, \mathbf{G}\}^*$ is a proper intermediate logic, because LC is the logic of linearly ordered Kripke-frames.

PROPOSITION 2.1. *The logic* LC *is a proper intermediate logic.*

Minimal Formula We write $\leqslant_{\mathrm{sub}}$ for the minimal preorder on Fml satisfying $\alpha \leqslant_{\mathrm{sub}} [\beta/p]\alpha$. In other words,

$$\alpha \leqslant_{\mathrm{sub}} \beta \text{ iff } \beta \text{ is a substitution instance of } \alpha.$$

Also, we write $\alpha \equiv_{\boldsymbol{\alpha}} \beta$ if $\alpha \leqslant_{\mathrm{sub}} \beta$ and $\beta \leqslant_{\mathrm{sub}} \alpha$. Note that, $\alpha \equiv_{\boldsymbol{\alpha}} \beta$ if and only if β can be obtained from α by renaming propositional variables, and vice versa. Also, we write $<_{\mathrm{sub}}$ for the binary relation $(\leqslant_{\mathrm{sub}}) \backslash (\equiv_{\boldsymbol{\alpha}})$.

Example 2.2. For example, $p \to q <_{\mathrm{sub}} p \to (p \to r)$. It is because $[p \to r/q](p \to q) = p \to (p \to r)$ and one can see that $p \to (p \to r) \not\leqslant_{\mathrm{sub}} p \to q$.

Let L be a logic. A formula α is *minimal* in L if α is a minimal element of the preordered set $\langle L, \leqslant_{\mathrm{sub}} \rangle$ (i.e., $\alpha \in L$ and there does not exist a formula $\beta \in L$ such that $\beta <_{\mathrm{sub}} \alpha$). Trivially, it is decidable whether a given formula α is minimal in L if the logic L (i.e., "$\alpha \in L$?") is decidable. It is because for each formula α, the number of formulas $\beta \leqslant_{\mathrm{sub}} \alpha$ is bounded up to $\equiv_{\boldsymbol{\alpha}}$.

Example 2.3. $\mathbf{K}$, $\mathbf{S}$, and $\mathbf{P}$ are minimal in CL.

Example 2.4. $\mathbf{G}$ is not minimal in CL, because $((a \to b) \to c) \to ((b \to d) \to c) \to c <_{\mathrm{sub}} \mathbf{G}$ and $((a \to b) \to c) \to ((b \to d) \to c) \to c \in \mathrm{CL}$.

3. The Komori–Kashima Problem

In this section, we restate below the statement of the Komori–Kashima problem, and then we present a few observations.

Problem. (the Komori–Kashima problem [8,10] (restated)) The implicational intermediate logics axiomatizable by formulas minimal in CL are only IL and CL? In other words, there does not exist a proper implicational intermediate logic which is axiomatizable by formulas minimal in CL?

In the following, we present a few observations for this problem.

3.1. A Simplification of the Checking of the Minimality

To check the minimality is cumbersome, whereas it is decidable. In this subsection, we give a relatively simplified method for checking the minimality in Proposition 3.2. We first remark the following.

Proposition 3.1. *Let $<$ be the minimal strict preorder satisfying $\alpha < [\beta/p]\alpha$, where (1) $\beta = q$ for some $q \neq p$ occurring in α or (2) $\beta = q \to r$ for some distinct q and r not occurring in α. Then, $<$ coincides with the substitution strict preorder $<_{\mathrm{sub}}$.*

Proof. (Proof Sketch) The direction $(<_{\mathrm{sub}}) \subseteq (<)$ is only non-trivial. It suffices to show that if $\alpha <_{\mathrm{sub}} [\beta/p]\alpha$, then $\alpha < [\beta/p]\alpha$. For example, let $\beta \triangleq p_1 \to p_2 \to p_3$. Then $\alpha < [\beta/p]\alpha$ is shown by $[\beta/p]\alpha = [p_1/q_1][p_2/q_2][p_3/q_3][q_2 \to q_3/r][q_1 \to r/p]\alpha$, where q_1, q_2, q_3, r are pairwise distinct propositional variables not occurring in α. For general cases, it can be shown in the same manner. ∎

We say that a formula α is *not simple* if there exist $\beta <_{\mathrm{sub}} \alpha$, p, q, and r such that $\alpha = [q \to r/p]\beta$, where q and r are distinct variables not occurring in β. We say that α is *simple* otherwise. For example, the formula $(a \to b) \to (a \to b)$ $(= [a \to b/c](c \to c))$ is not simple, but the formula $c \to c$ is simple.

For simple formulas, we can check the minimality more easily, in that it suffices to consider only substitutions of a *variable*.

Proposition 3.2. *For each simple formula $\alpha \in L$, α is* minimal *in L if and only if for every formula $\beta <_{\mathrm{sub}} \alpha$ such that $[q/p]\beta = \alpha$ for some variables p and q occurring in β, $\beta \notin L$.*

Proof. By Proposition 3.1 and the definition of simple formula. ∎

Finally, we remark that simple formulas are sufficient to consider the axiomatization of intermediate logics.

PROPOSITION 3.3. *Every implicational intermediate logic is axiomatizable by simple formulas.*

PROOF. (PROOF SKETCH) It suffices to show that $L + [q \to r/p]\beta = L + \beta$, where q and r are distinct variables not occurring in β. $[q \to r/p]\beta \in L + \beta$ is trivial. $\beta \in L + [q \to r/p]\beta$ (i.e., $\beta \notin L \Rightarrow [q \to r/p]\beta \notin L$) can be easily shown by using Heyting algebras and maps such that q maps to the top element. ■

3.2. On the Decidability of the Komori–Kashima Problem

We remark that it is decidable whether a given formula α minimal in CL is a counter-example to the Komori–Kashima problem (i.e., $\mathrm{IL} \subsetneq \mathrm{IL}+\alpha \subsetneq \mathrm{CL}$?). It is because $\mathrm{IL} \subsetneq \mathrm{IL} + \alpha \subsetneq \mathrm{CL}$ if and only if the following two decidable conditions hold (see [5, Theorem 1]):

- $\alpha \notin \mathrm{IL}$,
- α is valid in the three-valued (linear) Heyting algebra.

Remark 3.4. Thus, it is also indeed possible to construct a semi-algorithm to find a counter-example by enumerating all formulas. However, we could not find any counter-example in this approach, so there would not exist a counter-example of a very small size.

The counter-example presented in the next section is actually obtained by an adhoc construction.

4. A Counter-Example

In this section, we give a counter-example formula to the Komori–Kashima problem. We show that this formula axiomatizes a proper intermediate logic. The goal to this section is to show the following.

THEOREM 4.1. (Main Theorem) *There exists a proper intermediate logic axiomatizable by formulas minimal in* CL. *More concretely, the logic* LC *is axiomatizable by formulas minimal in* CL.

To prove Theorem 4.1, let consider the formula $\mathbf{G}'$ defined as follows:

$$\begin{aligned}\mathbf{G}' \triangleq\ & (a \to a') \to (b' \to b) \to (c'' \to c') \to (c''' \to c') \to \\ & ((a \to b) \to c'') \to ((b' \to a') \to c''') \to \\ & (((a' \to b') \to c'') \to ((b \to a) \to c''') \to c' \to c) \to c.\end{aligned}$$

Then it suffices to show the following lemma.

LEMMA 4.2. *(1)* $\mathrm{IL} + \mathbf{G}' = \mathrm{LC}$.
(2) $\mathbf{G}'$ *is minimal in* CL.

PROOF OF LEMMA 4.2(1). Since $\mathrm{LC} = \mathrm{IL} + \mathbf{G}$, it suffices to show that both $\mathbf{G} \in \mathrm{IL} + \mathbf{G}'$ and $\mathbf{G}' \in \mathrm{IL} + \mathbf{G}$ hold. $\mathbf{G} \in \mathrm{IL} + \mathbf{G}'$ is shown by $(\star 1)$: $[a\ a\ b\ b\ c\ c\ c/a\ a'\ b\ b'\ c\ c'\ c''\ c''']\mathbf{G}' \to \mathbf{G} \in \mathrm{IL}$. $\mathbf{G}' \in \mathrm{IL} + \mathbf{G}$ is shown by $(\star 2)$: $\mathbf{G} \to \mathbf{G}' \in \mathrm{IL}$. Each proof of $(\star 1)$ and $(\star 2)$ is implemented in Coq [14]; see the supplement file "G.v" in [13]. See "Appendix 5.3" for simply-typed lambda terms for $(\star 1)$ and $(\star 2)$. ■

PROOF OF LEMMA 4.2(2). By Proposition 3.2 (since $\mathbf{G}'$ is simple), It suffices to check that all the $3+3+3+3+1+3+3+3 = 22$ formulas obtained by renaming some occurrences of a variable among $a, a', b, b', c, c', c'', c'''$ are not in CL. For example,

$$\begin{aligned}\mathbf{G}'_{\mathbf{11}} \triangleq\ & (X \to a') \to (b' \to b) \to (c'' \to c') \to (c''' \to c') \to \\ & ((a \to b) \to c'') \to ((b' \to a') \to c''') \to \\ & (((a' \to b') \to c'') \to ((b \to a) \to c''') \to c' \to c) \to c.\end{aligned}$$

is one of the 22 formulas since $\mathbf{G} = \mathbf{G}'_{\mathbf{11}}[a/X]$. This formula is not in CL, which can be shown by using the map such that a, b, b', c', c'' maps to *true* and X, a', c, c''' maps to *false*. Similarly for the other 21 formulas. It is straightforward, but is cumbersome. A proof is presented by using Coq; see the supplement file "G_minimal.v" in [13]. One can see that the tactic tauto [14, p. 310], that is an implementation of a complete decision procedure for intuitionistic propositional calculus (so, also applicative for classical propositional calculus) based on [3], fails for each of all the 22 formulas. ■

By this lemma, Theorem 4.1 has been proved. Hence, there exists a counter-example to the Komori–Kashima problem.

5. Conclusion and Open Problems

In this paper, we have shown that there exists a counter-example to the Komori–Kashima problem. In the following, we remain some open problems.

5.1. Open Problem 1

We revealed that there exists a formula minimal in CL that axiomatizes a proper intermediate logic (precisely, LC). However, the following problem remains open.

Open Problem 1. How many intermediate logics axiomatizable by formulas minimal in CL? Can we characterize the class of such logics?

5.2. Open Problem 2

The *order* [1] of a formula α, $\mathrm{Ord}(\alpha)$, is defined as follows:

$$\mathrm{Ord}(p) = 1, \qquad \mathrm{Ord}(\alpha \to \beta) = \max\{\mathrm{Ord}(\alpha) + 1, \mathrm{Ord}(\beta)\}.$$

For example, $\mathrm{Ord}(\mathbf{K}) = 2$, $\mathrm{Ord}(\mathbf{S}) = 3$, and $\mathrm{Ord}(\mathbf{P}) = 4$.

Corollary 5.1. (of Theorem 4.1) *There exists a proper intermediate logic axiomatizable by formulas minimal in* CL *and of order at most* 5.

However, it remains open whether or not there exists a formula of order 4 which defines a proper intermediate logic.

Open Problem 2. Does there exist a proper intermediate logic axiomatizable by formulas minimal in CL and of order at most 4? (Or equivalently, whether or not does the Komori–Kashima conjecture hold for formulas of order at most 4?)

5.3. Open Problem 3

In 2005, Komori [11] also posed the following problem.

Open Problem 3 ([11]). If possible, find an independent axiom system of implicational formulas minimal in CL which is obtained from the system $\mathbf{BCIK'W'}$ by adding formulas as axioms. Here,

$$\begin{aligned}
\mathbf{B} &\triangleq (b \to c) \to (a \to b) \to a \to c,\\
\mathbf{C} &\triangleq (a \to b \to c) \to b \to a \to c,\\
\mathbf{I} &\triangleq a \to a,\\
\mathbf{K'} &\triangleq (a \to b) \to a \to c \to b,\\
\mathbf{W'} &\triangleq (a \to a \to a \to b) \to a \to b.
\end{aligned}$$

Note that these five formulas are minimal in CL and $\{\mathbf{B}, \mathbf{C}, \mathbf{I}, \mathbf{K'}, \mathbf{W'}\}^* =$ IL. For example, the system $\mathbf{BCIK'W'P}$ is not independent, because $\mathbf{W'}$ is proved from $\mathbf{BCIP}$ [11]. Note that if the Komori–Kashima problem (precisely, the Komori–Kashima problem for super-$\mathbf{BCK}$ logics in the footnote[1] of the present paper) was positively solved, then there does not exist such

an independent axiom system (this is remarked in [12, Sect. 4]). As a consequence of the present paper, the Komori–Kashima problem for super-intuitionistic logics (hence for super-**BCK** logics) is negatively solved. However, the open problem above remains open. In particular, it is also open whether the system $\mathbf{BCIK'W'G'}$ (or a variant of the system) is an *independent* axiom system or not. (Of course, since $\mathbf{G'}$ is minimal in CL, if this is positively solved, the open problem above is also positively solved.)

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A. Supplement for Lemma 4.2(1)

In this short section, we explicitly present simply-typed lambda terms for proving ($\star 1$) and ($\star 2$) in Lemma 4.2(1), where the implicit brackets are right associative for application (see also the supplement file "G.v" in [13]).

($\star 1$): $[a\ a\ b\ b\ c\ c\ c\ c/a\ a'\ b\ b'\ c\ c'\ c''\ c''']\mathbf{G'} \in \mathrm{IL} + \mathbf{G}$ can be shown by the proof corresponding to the following simply-typed lambda term:

$$
\begin{aligned}
&\lambda x_1^{a\to a'}.\lambda x_2^{b'\to b}.\lambda x_3^{c''\to c'}.\lambda x_4^{c'''\to c'}.\lambda x_5^{(a\to b)\to c''}.\lambda x_6^{(b'\to a')\to c'''}.\\
&\lambda x_7^{((a'\to b')\to c'')\to((b\to a)\to c''')\to c'\to c}.\\
&\quad(\lambda x_8^{a\to b}.\\
&\qquad(\lambda x^{a'\to b'}.x_8x_5)\\
&\qquad(\lambda x_9^{b\to a}.(\lambda x_{10}^{b'}.((x_{10}x_2)x_9)x_1)x_6)\\
&\qquad((x_8x_5)x_3)\\
&\qquad x_7)\\
&\quad(\lambda x_8^{b\to a}.\\
&\qquad(\lambda x_9^{a'\to b'}.(\lambda x_{10}^{a}.((x_{10}x_1)x_9)x_2)x_5)\\
&\qquad(\lambda x^{b\to a}.(\lambda x_9^{b'}.((x_9x_2)x_8)x_1)x_6)
\end{aligned}
$$

$$
\begin{array}{l}
(((\lambda x_9^{b'}.(((x_9x_2)x_8)x_1)x_6)x_4) \\
x_7) \\
t^{\mathbf{G}}
\end{array}
$$

Here, $t^{\mathbf{G}}$ is a term having the type $\mathbf{G}$. Similarly, $(\star 2)$: $\mathbf{G} \in \mathrm{IL} + \mathbf{G}'$ can be shown by the proof corresponding to the following simply-typed lambda term:

$$
\begin{array}{l}
\lambda x_1^{(a\to b)\to c}.\lambda x_2^{(b\to a)\to c}. \\
\quad (\lambda x_3^a.x_3) \\
\quad (\lambda x_3^b.x_3) \\
\quad (\lambda x_3^c.x_3) \\
\quad (\lambda x_3^c.x_3) \\
\quad x_1 \\
\quad x_2 \\
\quad (\lambda x^{(a\to b)\to c}.\lambda x^{(b\to a)\to c}.\lambda x_3^c.x_3) \\
\quad t^{\mathbf{G}'}
\end{array}
$$

Here, $t^{\mathbf{G}'}$ is a term having the type $\mathbf{G}'$.

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Y. Nakamura
Department of Computer Science
Tokyo Institute of Technology
2-12-1, Ookayama, Meguro-ku
Tokyo 152-8550
Japan
nakamura.yoshiki.ny@gmail.com

N. Matsuda
Research Institute for Programming Science
Kanagawa University
2946, Tsuchiya, Hiratsuka-shi
Kanagawa 259-1293
Japan
matsuda.naosuke@gmail.com